Course: BTech **Semester:** 6

Prerequisite: Fundamental of Programming, Computer Network | 203105255 - Computer Networks

Rationale : Cyber security is the application of technologies, processes, and controls to protect systems, networks, programs, devices, and data from cyber-attacks. It aims to reduce the risk of cyber-attacks and protect against the unauthorized exploitation of systems, networks, and technologies.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T -** Theory, **P –** Practical

List of Practical

1.	**Practical-1** Implementation to gather information from any PC’s connected to the LAN using whois, port scanners, network scanning, Angry IP scanners etc.
2.	**Practical-2** Experiments with open source firewall/proxy packages like iptables, squid etc.
3.	**Practical-3** Implementation of Steganography.
4.	**Practical-4** Implementation of MITM- attack using wireshark / network sniffers.
5.	**Practical-5** Implementation of Windows security using firewalls and other tools.
6.	**Practical-6** Implementation to identify web vulnerabilities, using OWASP project.
7.	**Practical-7** Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.
8.	**Practical-8** Implementation of OS hardening and RAM dump analysis to collect the Artifacts and other Information.
9.	**Practical-9** Implementation of Mobile Audit and generate the report of the existing Artifacts.
10.	**Practical-10** Implementation of Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery.

Course Outcome
After Learning the Course the students shall be able to:
After learning the course, the students should be able to: 1. Explain the features and characteristics of the Linux Operating System and Windows Operating System. 2. Apply network monitoring tools to identify attacks against network protocols and services. 3. Apply various methods to prevent malicious access to computer networks, hosts, and data. 4. Explain how to investigate endpoint vulnerabilities and attacks. 5. Analyze network intrusion data to verify potential exploits. 6. Apply incident response models to manage network security incidents